Array Of Products

AlgoExpert · Medium · Arrays

The problem

- Out[i] = product of every element except a[i], without division.
- Prefix products then suffix products.

Intuition

- Each output is the product of everything to the left times everything to the right, so two sweeps replace division.

Brute force vs optimal

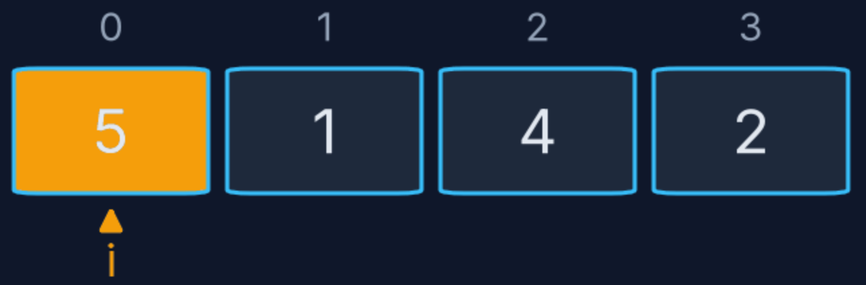
- Brute force: for each i, multiply all others → $O(n^2)$
- Total product ÷ a[i] → O(n), but breaks on zeros (and is disallowed)
- Prefix × suffix products → O(n), no division

Approach

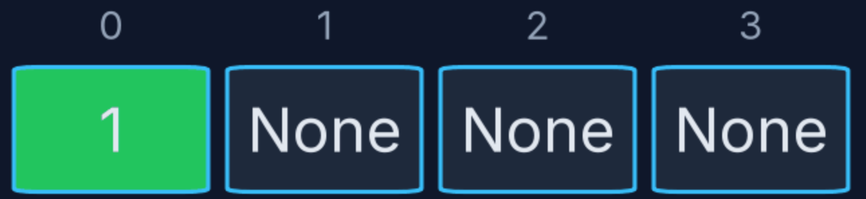
- 1. Fill each slot with the running product from the left
- 2. Sweep from the right with a second running product
- 3. Multiply it in
- 4. No division needed

Worked example

input



output after left pass

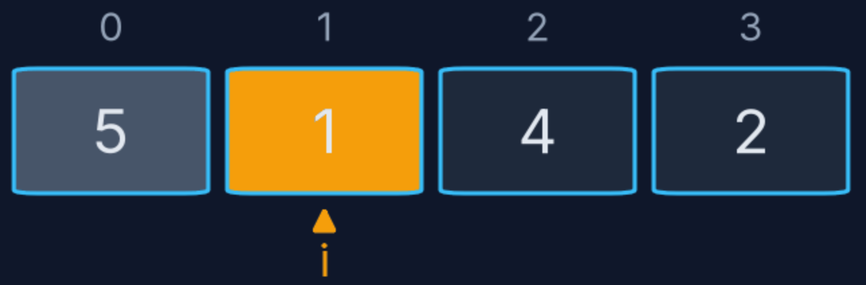



```
left product before 0 = 1
```

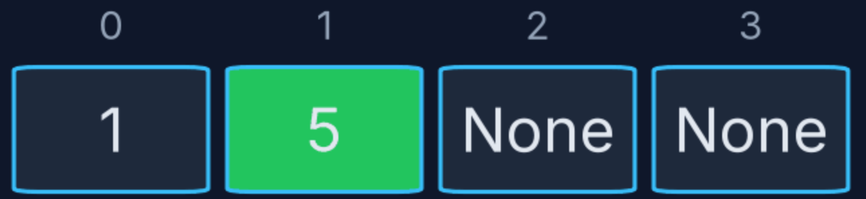
Left pass: out[0] = product of everything left = 1

Worked example

input



output after left pass

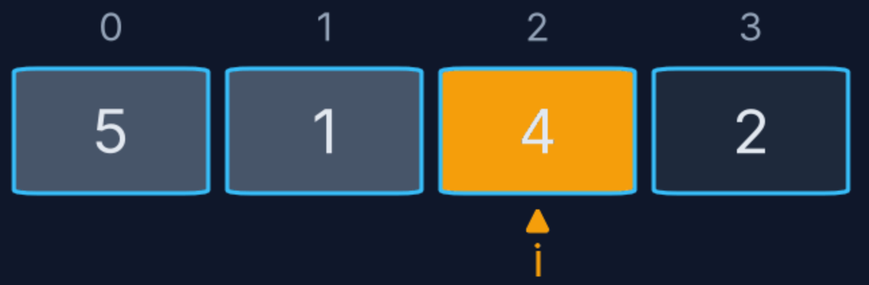



```
left product before 1 = 5
```

Left pass: out[1] = product of everything left = 5

Worked example

input



output after left pass

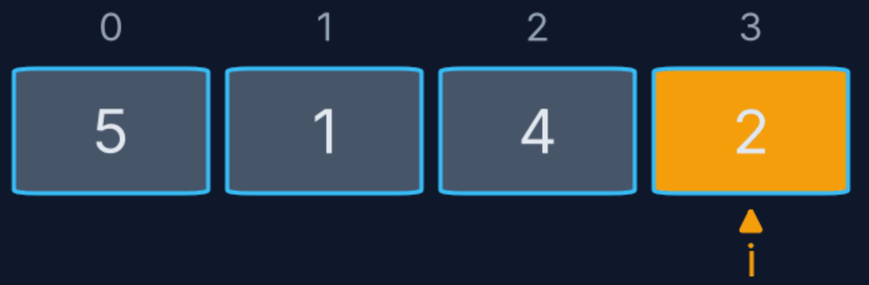



```
left product before 2 = 5
```

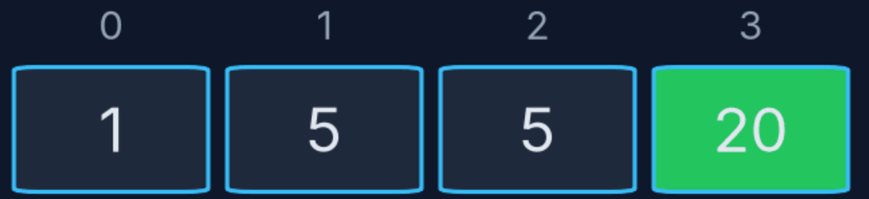
Left pass: out[2] = product of everything left = 5

Worked example

input



output after left pass

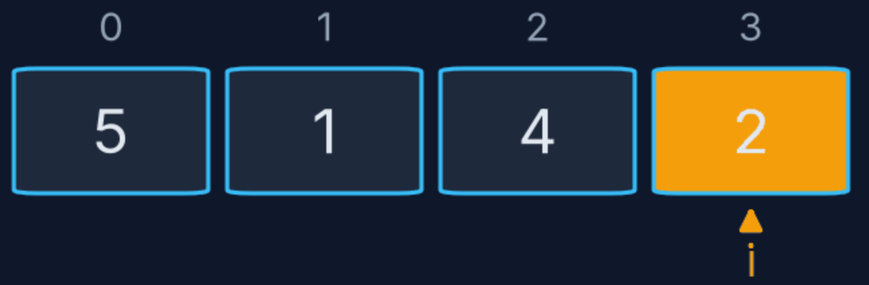



```
left product before 3 = 20
```

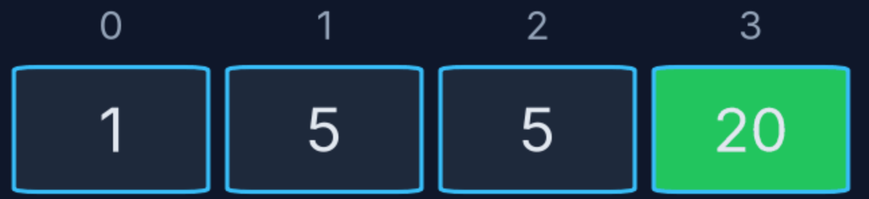
Left pass: out[3] = product of everything left = 20

Worked example

input



output after right pass

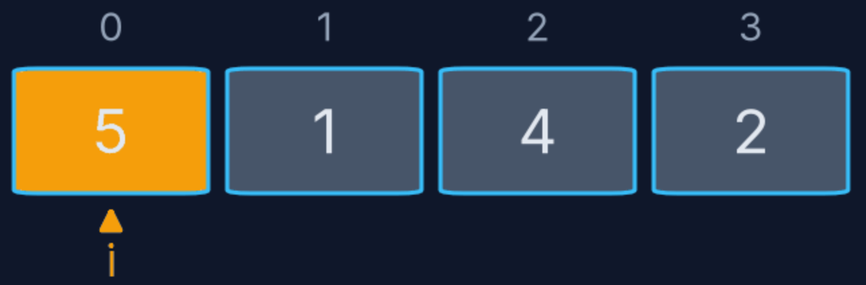



```
right product after 3 = 1
```

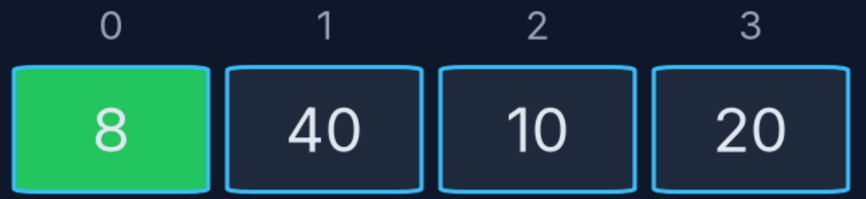
Right pass: out[3] ×= 1 → 20

Worked example

input



output after right pass




```
right product after 0 = 8
```

Right pass: out[0] ×= 8 → 8

Java solution

```java
public class ArrayOfProducts {
    public static int[] arrayOfProducts(int[] a) {
        int[] out = new int[a.length];
        int run = 1;
        for (int i = 0; i < a.length; i++) { out[i] = run; run *= a[i]; }
        run = 1;
        for (int i = a.length - 1; i >= 0; i--) { out[i] *= run; run *= a[i]; }
        return out;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n) output

Edge cases & pitfalls

- Zeros: one zero → only its index is non-zero
- Two zeros → everything is 0
- Overflow for large products

Interview tips & follow-ups

- Mention the division idea and why it fails
- Follow-up: O(1) extra space (reuse the output array)

Recap

- Each output is the product of everything to the left times everything to the right, so two sweeps replace division.
- Complexity: O(n)